

A Differentiable Atari VCS

A Complex, Fully Known Ground Truth for Explainable AI

Anonymous Submission

AAAI-27 — Supplementary Video

A real computer architecture, made fully differentiable

Two ports, bit-exact on all 64 Atari games

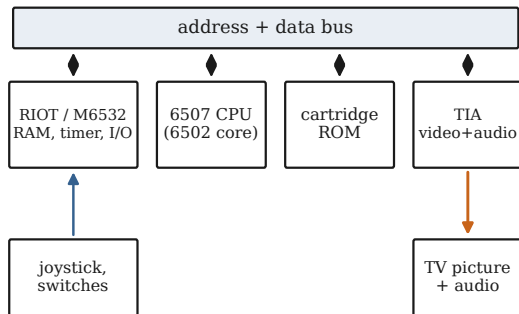
Gradients with a ground truth

Explanation needs ground truth

- ▶ To check an explanation, we must know the true mechanism
- ▶ Simple models: mechanism known, but explaining it tests nothing
- ▶ Deep networks: explanation needed, but no inner ground truth
- ▶ So an explanation can be confident and wrong
- ▶ We want a system that is complex, fully known, and differentiable

A study object: the Atari 2600

Atari 2600 VCS



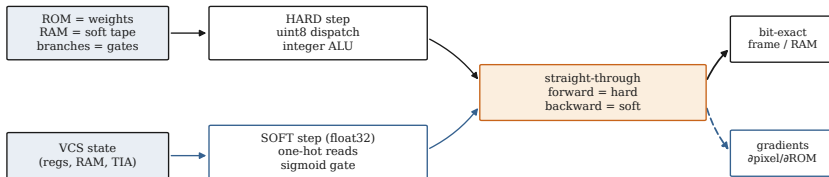
- ▶ A real 1977 computer: CPU, RAM, graphics chip, cartridge
- ▶ The platform that launched deep reinforcement learning
- ▶ Complex, yet fully specified to the last bit
- ▶ We re-express its every step so gradients can flow

Two independent differentiable ports

	JUTARI	JAXTARI
Language / AD	Julia / Zygote	JAX / XLA
RAM-exact (of 64)	64	64
Pixel-exact (of 64)	64	64
Single env, CPU	fastest	vmap-batched
Batched on GPU	—	~3M steps/s

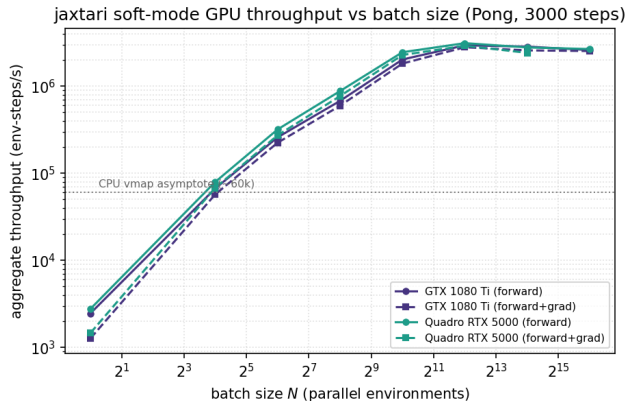
- ▶ Built twice, in two languages, against one reference
- ▶ Both match XITARI bit-for-bit on all 64 games

Soft equals hard



- ▶ ROM as a weight tensor, RAM as a soft tape, branches as gates
- ▶ Forward pass is bit-exact equal to the hard machine
- ▶ Backward pass gives surrogate gradients where bits have none

Throughput: a GPU path



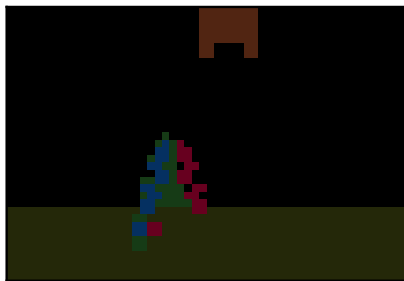
- ▶ JUTARI: fastest per environment on CPU
- ▶ JAXTARI: batches the rollout on a GPU
- ▶ Millions of environment-steps per second
- ▶ Gradients cost only a few percent more

Gradients with a ground truth

(a) real SI, 35 s



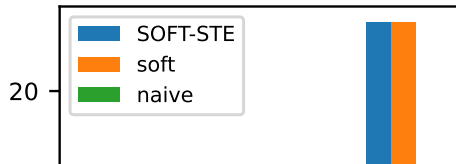
(b) $\partial \text{screen} / \partial \text{RIGHT}$ (sampler)



(c) naive $\equiv 0$ (vanishes)



(d) inverse $\partial(\text{move right}) / \partial \text{joy}$



In the supplementary material

- ▶ Full proofs of the forward-equivalence and temperature-limit results
- ▶ A study of how the relaxation parameters shape the gradient
- ▶ Every game, with its ROM hash, mapper, and per-frame exactness
- ▶ Throughput tables for CPU and GPU
- ▶ These side-by-side comparison videos

A testbed for explainable AI

- ▶ A complex system that is fully known and fully differentiable
- ▶ Built twice, validated bit-for-bit on all 64 games
- ▶ Exact frames, with surrogate gradients for attribution
- ▶ The XAI ground truth we hope the community takes up

Full code of both ports released under the MIT license upon acceptance.
Thank you.